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February 2, 1981
SUPERSEDING
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FEDERAL SPECIFICATION

SEALERS, SANDING, LACQUER-TYPE (FOR WOOD FURNITURE)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 Scope. This specification covers a sanding sealer of the lacquer type suitable for spray application (conventional or airless) on wood furniture.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-C-96 - Cans, Metal, 28-Gage and Lighter.

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and commercial item descriptions, as outlined under General Information in the Index of Federal Specifications, Standards, and Commercial Item Descriptions. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification, other Federal specifications, and commercial item descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston; New York; Philadelphia; Washington, DC; Atlanta; Chicago; Kansas City, MO; Fort Worth; Houston; Denver; San Francisco; Los Angeles; and Seattle, WA.

(Federal Government activities may obtain copies of Federal specifications, standards, and commercial item descriptions, and the Index of Federal Specifications, Standards, and Commercial Item Descriptions from established distribution points in their agencies.)

Military Standard:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military specifications and standards required by contractor in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standards.

D1544 - Color of Transparent Liquids (Gardner Color Scale).
D1545 - Viscosity of Transparent Liquids by Bubble Time Method.
D1644 - Nonvolatile Content of Varnishes.
D3133 - Quantitative Determination of Cellulose Nitrate in Alkyd Lacquers by Infrared Spectrophotometry.
D3272 - Vacuum Distillation of Solvents from Solvent Base Paints for Analysis.
D3280 - White Zinc Pigments.
D3335 - Low Concentrations of Lead and Cadmium in Paint by Atomic Absorption Spectroscopy.
E 260 - Recommended Practice for General Gas Chromatography Procedures.

(Application for copies should be address to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Volatile solvent. The solvent, when tested as specified in 4.4.1, shall conform to the following requirements by volume:

(a) The total of solvents with olefinic or cyclo-olefinic unsaturation shall not exceed 5 percent.
(b) The total of aromatic compounds with eight or more carbon atoms in the molecule, except ethylbenzene, methylbenzoate, and phenyl acetate, shall not exceed 8 percent.
(c) The total of ethylbenzene, toluene, and branched-chain ketones shall not exceed 20 percent.
(d) A solvent which may be classified into more than one of the above groups shall be considered a member of the group having the lowest allowable concentration.
(e) The total of (a), (b), and (c) shall not exceed 20 percent.
(f) Halogenated solvents shall not be present.
(g) Methyl n-butyl ketone shall not be present.

3.2 Quantitative requirements.

3.2.1 The quantitative requirements of the sealer shall be as specified in table I.

TABLE I. Quantitative requirements

Characteristics	Requirements	
	Minimum	Maximum
Nonvolatile matter, percent by weight of sealer	21	—
Nitrocellulose, percent by weight of nonvolatile matter	40	—
Zinc stearate, percent by weight of nonvolatile matter	4	6
Viscosity, Gardner-Tubes	A	C
Dry time, minutes	—	20
Lead content, percent by weight of nonvolatile	—	0.06

3.3 Qualitative requirements.

3.3.1 Condition in container. The sealer shall be free from grit, seeds, skins, lumps, and livering when tested as specified in 4.4.10.

3.3.2 Storage stability (partially full container). The sealer shall show no skinning when tested as specified in 4.4.2.

3.3.3 Color. When tested as specified in 4.4.3, the color shall not be darker than Gardner Color Standard No. 7.

3.3.4 Spraying properties. The sealer shall not run, sag, streak, or show orange peel when tested as specified in 4.4.4.

3.3.5 Sanding properties. When tested as specified in 4.4.5, the sealer shall sand down to a smooth, uniform film, and the sanded-off portion shall be in the form of a dry powder. The sandpaper shall show no gumming.

3.3.6 Compatibility. When tested as specified in 4.4.8, the sealer shall form a smooth, uniform film over the coating system, and shall not lift from the coating system, cause the coating system to lift from the wood, or cause the wood grain to be raised.

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3.4 Precautionary marking. The precautionary marking shall contain the following (see 5.2).

"Flammable, volatile solvent."
"Use with adequate ventilation."
"Avoid prolonged breathing of vapors or spray mists."

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein.

4.2 Inspection of preparation for delivery. An inspection shall be made to determine whether the packaging, packing, and marking comply with the requirements of section 5. The sample unit shall be one shipping container fully prepared for delivery and selected at random. Sampling shall be in accordance with MIL-STD-105. The inspection level shall be S-2 with an AQL of 4.0 expressed in terms of percent defective.

4.3 Testing of the end item.

4.3.1 Lot. The sealer shall be assembled into lots as specified in MIL-STD-105. In MIL-STD-105, the words "essentially the same conditions" shall be interpreted to mean a manufacturer's batch and defined as the end product of all raw materials mixed, blended, or processed in a single operation.

4.3.2 Sampling of the end item. For the purposes of sampling, the lot shall be expressed in units of gallons. Samples from lots shall be taken in accordance with MIL-STD-105 using inspection level S-2 and an acceptable quality level (AQL) of 1.5.

4.4 Test procedures. The sealer shall be tested according to the test methods indicated in table II. Unless otherwise specified, all tests shall be performed at standard testing conditions which are a temperature of $23^{\circ} \pm 1^{\circ}$ C and a relative humidity of 50 ± 5 percent. All test reports shall contain the individual values utilized in expressing the final result. All tests shall be evaluated for conformance to the requirements specified in section 3. Failure to pass any test, or noncompliance with any requirement, shall be cause for rejection of the sample.

TABLE II.

Characteristics	Required referencer	Applicable Test Method	
		ASTM Method	Paragraph reference
Solvent	3.1	D3272	4.4.1
Condition in container	3.3.1	-------	4.4.10
Storage stability:			
Partially full container	3.3.2	-------	4.4.2
Color	3.3.3	D1544	4.4.3
Spraying properties	3.3.4	-------	4.4.4
Sanding properties	3.3.5	-------	4.4.5
Compatibility	3.3.6	-------	4.4.8
Nonvolatile matter	Table I	D1644	-----
Nitrocellulose	Table I	D3133	-----
Zinc stearate	Table I	D3280	4.4.6
Viscosity	Table I	D1545	-----
Dry time	Table I	-------	4.4.7
Lead content	Table I	-------	4.4.9

4.4.1 Solvent analysis. Analyze the solvent by gas chromatography. The solvent shall be separated from the sealer as specified in ASTM D 3272 and then injected into the gas chromatograph, or the sealer shall be injected directly using equipment specifically designed for that purpose. The procedure specified in ASTM E 260, and any apparatus, operating conditions, columns, and options permitted herein shall be used for the chromatographic analysis. The accuracy of the analysis shall be 0.25 percent absolute by weight or less for each component and the reproducibility shall be 0.25 percent absolute by weight over three or more runs. All peaks 0.5 percent of the sample or greater shall be identified and quantified. Convert percent by weight to percent by volume, and evaluate the results for compliance with 3.1.

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4.4.2 Storage stability (partially full container). A 6-ounce sample of the sealer shall be measured into an 8-ounce wide-mouth glass jar, measuring 114 mm (4-1/2 in) in height and 51 mm (2 in) in diameter. The cover shall be secured on tightly and the jar inverted momentarily. The jar shall be placed in an upright position in the dark for 48 hours at standard conditions. The sample shall not be agitated or disturbed. Examine after 48 hours for compliance with 3.3.2.

4.4.3 Color. The color shall be determined on the clear liquid after centrifuging out the sanding agent in accordance with ASTM D 1544 for compliance with the requirement in table I.

4.4.4 Spraying properties. A glass panel with no side shorter than 102 mm (4 in) and a minimum area of 206 cm^2 (32 in^2) shall be cleaned to no water break. The sealer shall be reduced in accordance with the manufacturer's directions. Spray one cross-coat of the sealer on the prepared panel, holding the gun perpendicular to the panel at a distance of 203 mm (8 in) and moving in a straight line across the face of the panel. Immediately place the panel in a nearly vertical position and allow to dry at standard conditions for 1 hour. Evaluate the dried film for compliance with 3.3.4.

4.4.5 Sanding properties. Sand a test panel prepared in 4.4.4 with unused No. 5/0 or finer sandpaper, employing the following technique: Attach the finishing paper to one face of a metal or weighted wood sanding block, which shall be approximately 13 cm^2 (2 in) having a total weight of 1200 ± 5 grams, and have the edges and corners of the bearing face rounded off. The finishing paper shall completely cover and fit tightly against the face of the block, so as to avoid the formation of wrinkles or creases in the paper while being used. Any sharp folds or edges in the paper around the edge of the block shall be bruised, softened, or feathered before using. Place the paper-covered end of the block on the coating. Grip the block firmly so that it may be slowly moved back and forth across the coating in a straight line. Care should be taken that no downward pressure is exerted by the hand upon the block as the total vertical force exerted against the coating should be not more than the weight of the block (approximately 47 g/cm^2).

A completed movement of the block forward and back the full length of width of the panel shall be considered one cycle. Move the weight by hand over the coating the full length and width of the panel for a total of 25 cycles, changing the direction of movement lengthwise to crosswise and vice versa of the weight every tenth cycle. Evaluate the sanded coating for compliance with the requirement of 3.3.5.

4.4.6 Zinc stearate (sanding agent). Evaporate 10g of sealer on a steam bath, and ignite the residue in order to burn off all the organic matter. From the ash, determine zinc oxide content in accordance with ASTM D 3280. Convert the value obtained for zinc oxide to the corresponding value for commercial-grade zinc stearate by multiplying by 5.9. Evaluate the result for compliance with the requirement in table I.

4.4.7 Dry time. A glass panel shall be cleaned to no water break. Apply the sealer to the test panel at a dry film thickness of 25.4 µm (0.001 in). Place the test panel in a horizontal position at a height such that, when the thumb is placed on the film, the arm of the operator is in a vertical straight line from wrist to shoulder. Bear downward on the film with the thumb, exerting the maximum pressure of the arm, at the same time turning the thumb through an angle 90° in the plane of the film. The film shall be considered dry-through when no loosening, detachment, wrinkling, or other distortion of the film occurs. Determine compliance with the requirement in table I.

4.4.8 Compatibility.

4.4.8.1 Panel preparation. The panel shall be cut from synthetic resin-bonded, straight-grain, 5-ply walnut veneer (faces and backs) having no sizing or other surface pretreatment. They shall be approximately 8 mm (5/16 in) thick; no side of the panel shall be shorter than 102 mm (4 in), and the minimum area shall be 206 cm^2 (32 in^2).

The panel shall have a moisture content of not less than 6 percent, and not more than 8 percent at the time of test coating and sealer application. Panels shall be conditioned to a moisture content of 7 to 8 percent at 18 to 24°C (65° to 75°) and 35 to 40 percent relative humidity.

Prior to finishing, the panel shall be sanded smooth with 5/0 or finer sandpaper. Surfaces shall be thoroughly cleaned immediately before staining.

4.4.8.2 Application of coating system.

4.4.8.2.1 Stain. A non-grain-raising "fast-to-light" solvent-type stain shall be applied and allowed to dry thoroughly (not less than 30 minutes).

4.4.8.2.2 Washcoat. A washcoat consisting of a blend of seven parts denatured alcohol and one part refined, bleached shellac shall be applied. After the washcoat has thoroughly dried (not less than 30 minutes), the surface shall be scuff sanded with 5/0 or finer sandpaper, and all loose particles and dust removed.

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4.4.8.2.3 Filler. A paste wood filler shall be applied. The filler shall be thinned with a suitable solvent, such as low-flash naphtha, to increase its workability. When the filler has dried to a proper wiping consistency, it shall be worked into the pores by any means so that all pores shall be completely filled. Excess filler shall be thoroughly wiped from the panel. The filler shall be allowed to air dry for not less than 18 hours.

4.4.8.2.4 Sealer. The sealer shall be reduced in accordance with the manufacturer's directions. Spray one cross-coat of the sealer on the prepared panel, holding the gun perpendicular to the panel at a distance of 203 mm (8 in) and moving in a straight line acrosss the face of the panel. Immediately place the panel in a nearly vertical position and allow to dry at standard conditions for 1 hour. Evaluate the dried film for compliance with 3.3.6.

4.4.9 Lead content. The method described in 4.4.9.1 or the method described in 4.4.9.2, shall be used. The X-ray fluorescence spectrometry method described in 4.4.9.1 shall be the final determinant of compliance in all cases.

4.4.9.1 Analysis for lead by X-ray fluorescence spectrometry. Lead content shall be determined using an X-ray fluorescence spectrometer capable of determining lead content at a minimum level of 0.03 percent by weight. The molybdenum X-ray source shall be operated at 60 Kv and 45 Ma; a lithium fluoride crystal cut along the 200 planes shall be used to disperse fluorescent X-rays; the detector collimeter shall be set at fine; and flow and scintillation counters shall be used as detectors. Pulse height selection shall be used in all measurements, and the counting time shall be 100 seconds. Place the sample disc in the sample holder, and measure the count rates of lead, lead background, and the molybdenum Rayleigh-scattered background from the X-ray tube at the following angles:

Analytical Line	Angle	Intensity
Mo	20.33	I_{Mo}
Background I	33.50	I'
Pb	33.93	I_{Pb}
Background II	34.50	I"

4.4.9.1.1 Calculation. Determine the ratio R of the lead and molybdenum lines as follows:

$$R = \frac{2I_{Pb} - I' - I"}{2I_{Mo}}$$

4.4.9.1.2 Procedure. Duplicate drawdowns of the well-mixed sealer shall be made on the sealed portion of a standard sealed penetration chart, using a mechanical applicator plate and film applicator with a 150 μm gap. The drawdowns shall be at least 250 mm long, and shall be allowed to air-dry for 24 hours at standard conditions. Discs shall then be cut from each drawdown to fit the sample holder of an X-ray fluorescence spectrometer. Standard samples of the same type of paint containing known amounts of lead shall be prepared and measured in the same way, and the R values shall be plotted against the lead concentration to obtain a working curve. By use of the working curve, the value of R obtained for the test sample shall be converted to the lead concentration in percent by weight, and the result shall be evaluated for compliance with the requirement in table I.

4.4.9.2 Analysis for lead by Atomic Absorption Spectrometry. ASTM D3335 shall be used to determine the concentration of lead in percent by weight, and the result shall be evaluated for compliance with the requirement in table I.

4.4.10 Condition in container. Sealer in the original, unopened container shall be used in this test. The container shall not be agitated. After opening, note whether skinning has occurred and remove any continuous skin with a spatula by first carefully cutting the skin free from the container. Lower a spatula or paddle into the container and observe if the material is thickened or livered and if settling or caking exists. Thoroughly stir the sealer and check for undispersed lumps as it flows from the spatula or paddle. Flow some of the material on a glass panel, allow to drain in a vertical position, and examine for seeding and coarse particles. Evaluate for compliance with 3.3.1.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation-packaging shall be level A or commercial, as specified (see 6.2).

5.1.1 Level A. The 1-quart and 1-gallon quantities of sealer shall be packaged in metal cans conforming to PPP-C-96, type V, class 4. Exterior plan B coating and side seam striping shall be required. The cans shall be provided with handles which shall be galvanized or otherwise protectively coated to resist corrosion.

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See para. 5.2.1 for the preparation for delivery of 5-gallon and 55-gallon quantities.

5.1.2 Commercial. The 1-quart and 1-gallon quantities of sealer shall be packaged in accordance with normal commercial practice. The complete package shall be designed to protect the sealer against damage during shipment, handling and storage.

See para. 5.2.2 for the preparation for delivery of 5-gallon and 55-gallon quantities.

5.2 Packing. Packing shall be level A or commercial, as specified (see 6.2).

5.2.1 Level A. Twelve 1-quart cans or six 1-gallon cans of sealer, packaged as specified in 5.1, shall be packed in close-fitting boxes conforming to PPP-B-636, grade V3c, V3s, or V2s. The boxes shall be closed, waterproofed, and reinforced in accordance with the appendix to PPP-B-636. Alternatively, cleated plywood, wire bound or nailed wood boxes shall be acceptable shipping containers when lined with a waterproof barrier material. The edges of the barrier material shall be sealed with waterproof tape or adhesive. The 5-gallon and 55-gallon quantities of sealer shall be furnished in metal pails and metal drums, respectively, conforming to Item 260 of the National Motor Freight Classification and Rule 40 of the Uniform Freight Classification.

5.2.2 Commercial. The sealer shall be packed in a manner that will assure acceptance by common carrier and provide product protection against loss and damage during multiple shipments, handling and storage. The shipping container shall be in compliance with the National Motor Freight Classification and Uniform Freight Classification.

5.3 Marking. Marking shall be as specified in the contract or order (see 6.2).

5.4 Precautionary marking. Each container shall bear a precautionary marking conforming to 3.4.

6. NOTES

6.1 Intended use. The lacquer sealer covered by this specification is intended for use as a sanding sealer on wood furniture. It may be used in conjunction with a stain, washcoat, and filler, applied as specified in 4.4.8, together with lacquer finish coats.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Level of packaging and packing required (see 5.1).
(c) Size and type of container required (see 5.1).
(d) Marking required (see 5.3).

MILITARY INTERESTS:

DOD has waived coordination.

CIVIL AGENCY COORDINATING ACTIVITY:

NASA-JFK

Preparing activity:

GSA-FSS

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